

Vehicle Exterior Damage Segmentation: A Review of Convolutional, Transformer, and Hybrid Architectures for Automotive Damage Inspection

AutoInspect-X Capstone Literature Review

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Abstract—Automated visual inspection of vehicle exterior damage is a long-standing application of computer vision for insurance claims, used-vehicle inspection, and driving-assistance systems. The task has two dominant formulations — damage detection/classification and damage segmentation. A 2025 systematic review confirms that detection and classification have a large, saturated applied literature, whereas pixel-level segmentation of fine, spread-out damage remains an active front. This review synthesizes verified literature across four threads relevant to a bounded, single-photo damage segmentation study: (i) public damage-segmentation datasets (CarDD, VehiDE, CrashCar101 and related resources); (ii) convolutional segmentation architectures (FCN, U-Net, ResNet, DeepLab-style atrous designs, HRNet, Mask R-CNN) and efficient networks; (iii) transformer and CNN–transformer hybrid designs (ViT, SETR, SegFormer, Swin, TransUNet) and the role of attention for global context; and (iv) uncertainty/confidence estimation (aleatoric vs. epistemic taxonomy, MC dropout, deep ensembles) as the basis of an honesty contract between model predictions and claims. We close with the research gaps this evidence supports and clearly-separate literature findings from the preliminary experimental observations of the AutoInspect-X capstone, whose two 15-epoch pilots are explicitly not a conclusion. All cited works were verified against publisher, arXiv, or project pages in September 2026 unless flagged; no claim is strengthened beyond what the cited sources support.

Index Terms—vehicle damage, semantic segmentation, CNN, vision transformer, CNN–transformer hybrid, encoder–decoder, attention, mean IoU, uncertainty, confidence calibration.

I. Introduction

THE automatic assessment of vehicle exterior damage has been a practical computer vision problem for decades: an insurer photographing a claim, a used-car inspector, or a driver documenting an incident all need the same basic question answered — where is the damage and how bad is it? The commercial value is large: automated triage can route the simplest claims to low-cost settlement and free human assessors for complex cases. The most recent systematic review of the field screened 55 selected papers and grouped methods by task

(classification, detection, segmentation) and application context (insurance claim, traffic, post-accident) [1]. The picture it paints is consistent: damage detection (bounding boxes) and damage classification (scratch vs. dent vs. crack) are mature and saturated; damage segmentation (assigning a class label to each damaged pixel) is where the field is actively moving, because insurance and inspection applications need shape, extent, and location, not just a label. Commercial systems are closed-source and do not publish evaluation curves [1].

Yet the honest research value is bounded by data. Publicly available datasets with pixel-level damage annotations exist (CarDD [2], VehiDE [3], CrashCar101 [4]), but no public, licence-clear dataset carries observed repair costs; cost prediction therefore depends on private or synthetic sources and is not reproducible ground truth.

Three threads in the wider segmentation literature are directly relevant. First, convolutional encoder–decoder designs (U-Net [5]) remain the workhorse for small-data, fine-structure segmentation and are the ancestor of most automotive damage segmentation baselines. Second, transformer and CNN–transformer hybrid designs [6]–[10] add global context that plain convolutions struggle to obtain, which matters when a dent on a door and a displaced panel on an A-pillar are physically far apart yet causally linked. Third, uncertainty estimation [11]–[13] supplies the vocabulary for a system that labels its own confidence honestly instead of emitting raw softmax values as if they were truth.

This review serves a specific capstone — AutoInspect-X, a photo-first car damage segmentation demo with an explicit research question: does a CNN+transformer hybrid achieve higher segmentation quality than a plain U-Net baseline on CarDD under identical splits, seed, and schedule, and how honestly can the models label their own confidence? We restrict the review to verified, reproducible sources; claims are separated into what a source states, what is inferred from the evidence, and what

is a preliminary observation of the capstone’s own (single-seed, 15-epoch) pilots.

The review is organized as follows. Section II states the selection and integrity method. Section III defines the segmentation task and the metrics used to compare methods. Section IV reviews the verified damage datasets. Sections V–?? review the three architectural families (convolutional, transformer, hybrid). Section IX covers uncertainty and confidence. Section X synthesizes the gaps, and Section XIII separates literature findings from the capstone’s own preliminary observations.

II. Scope and Selection Method

We follow the selection criterion adopted in the AutoInspect-X research report: a work is included if it (i) defines the dataset, baseline, or training rule of this project; (ii) is a directly comparable architecture for the research question (transformer and hybrid segmentation); or (iii) supplies the uncertainty vocabulary that the honesty contract relies on. This is a targeted review for a bounded capstone, not an exhaustive survey; readers interested in a systematic landscape are directed to the systematic review of Hasan et al. [1].

Two integrity rules are enforced throughout. First, every citation was checked against publisher, arXiv, or project pages during the September 2026 revision of the project research report; where a previously-listed dataset could not be verified it was excluded (this applies to the unverified “CDD (2025)” and “Insurance-Damage-v2” names, which are not used as evidence here). Second, we never treat cross-paper numbers as comparable benchmarks: the studies differ in dataset version, split policy, pre-processing, and evaluation code, so their reported numbers are not directly comparable [1], [2].

III. Segmentation Task and Evaluation Metrics

A. Task formulations

Automotive damage analysis appears in four formulations that recur across the literature [1]:

- Classification — one label per image or per region (scratch, dent, crack, glass shatter, lamp broken, tire flat). Saturated.
- Detection — bounding boxes per damage instance. Mature.
- Semantic / instance segmentation — per-pixel class labels (semantic) or per-instance masks (instance). The active front [2]–[4].
- Salient object detection — separating “damaged region” from background without fine classes; supported by CarDD and VehiDE [2], [3].

AutoInspect-X targets semantic segmentation of damage classes from a single photo, which is the smallest reproducible formulation that still answers “where is the damage and how extensive”.

B. Metrics

The standard segmentation metrics descend from the PASCAL VOC evaluation protocol [14]. Mean intersection-over-union (mIoU) over classes is the primary metric in most damage-segmentation studies [2], [15]. For C classes, with ground-truth pixels G_c and predicted pixels P_c for class c :

$$\text{IoU}_c = \frac{|G_c \cap P_c|}{|G_c \cup P_c|}, \quad \text{mIoU} = \frac{1}{C} \sum_{c=1}^C \text{IoU}_c. \quad (1)$$

The Dice coefficient, closely related to IoU,

$$\text{Dice}_c = \frac{2|G_c \cap P_c|}{|G_c| + |P_c|}, \quad (2)$$

rewards overlap slightly differently (it equals the harmonic mean of precision and recall) and is often reported alongside IoU [2], [14]. Pixel accuracy — the fraction of all pixels correctly classified — is simple to compute but is a weak summary when the background dominates, which is always true in damage photos.

Four practical warnings about segmentation metrics recur in the literature [1], [14] and are adopted by AutoInspect-X:

- 1) Background dominance. If the background class is included in the mean, a large, correctly-predicted background can inflate mIoU; the AutoInspect-X protocol therefore reports foreground-only means, and absent classes score zero.
- 2) Class imbalance. Most damage classes are rare at the pixel level (a crack is a few thousand pixels in a 512×512 image). Mean metrics over classes therefore weigh rare classes equally with common ones, which is desirable but must be stated.
- 3) Small-damage slice. The binding constraint for trust in real claim photos is not the overall mean but the IoU restricted to the smallest ground-truth instances. The AutoInspect-X protocol defines a small-damage slice (damage instances at or below the 25th percentile of the training-set instance area) and reports slice IoU/Dice explicitly.
- 4) Split discipline. Cross-paper numbers are not comparable because papers differ in dataset version, split policy, pre-processing, and evaluation code [1]; within one study, the test split must be touched only once after model selection.

C. Task formulations in AutoInspect-X

A single-damage photo may contain several instances of several classes. The AutoInspect-X protocol uses the CarDD instance masks as real ground truth, collapses them to a per-pixel semantic label (background or one of the six classes) at 512×512 , and evaluates foreground mIoU (1), Dice (2), and pixel accuracy on the official train/val/test splits.

TABLE I
Verified public vehicle-damage datasets relevant to segmentation.

Dataset	Images	Annotation	Verified?
CarDD [2]	4,000	Polygon instance masks; 6 classes (dent, scratch, crack, glass shatter, lamp broken, tire flat)	Yes
VehiDE [3]	13,945	Detection, segmentation, salient object detection; 8 categories	Yes
CrashCar101 [4]	101,050	Synthetic mask; part + damage, severity	Yes
CDA-Net data [16]	11,380	Damage annotations, Indian market	Yes
Three-quarter-view [17]	n/a	Severity classification, Heliyon	Yes

IV. Vehicle Damage Datasets

Table I summarizes the verified public datasets used across damage analysis work that informed this review.

A. CarDD — working dataset of this project

CarDD (Wang, Li, Wu, 2023) is the first publicly available large-scale dataset for vision-based car damage detection and segmentation: 4,000 high-resolution images with over 9,000 annotated instance masks across the six damage classes above [2]. It supports object detection, instance segmentation, and salient object detection, and its primary reference model in the original paper is Mask R-CNN [2], [18]. For AutoInspect-X, CarDD is used with its official splits preserved (train 2,816 images / 6,211 instances, val 810 / 1,744, test 374 / 785 — measured from a local audit). CarDD contains damage masks only: no vehicle-part masks, no vehicle metadata, and no repair cost or repair-action annotations. This is why cost prediction is out of scope for the project: there is no public cost ground truth to learn from.

B. VehiDE

VehiDE [3] is a publicly released large-scale resource for segmenting and detecting automotive damage: 13,945 high-resolution photos with more than 32,000 annotated damage instances across eight damage categories. A follow-up study benchmarks segmentation methods on it [15].

An important verification note from the report: an earlier revision of the project documentation listed “VehiDE” as “12,000 images, 61 parts, 26 damage types”; that figure is not supported by the verified VehiDE numbers and instead matches the dataset described by the ALBERT model paper [19]. The 12,000/61/26 figure is treated as unverified and is not used as evidence.

C. CrashCar101 and other resources

CrashCar101 [4] is a procedurally generated dataset of 101,050 car-crash images with synthetic part and damage segmentation masks and structural severity levels. Because its imagery is synthetic, it complements real datasets as an augmentation or pre-training source, not as real-world validation evidence. CDA-Net [16] introduces an 11,380-image damage dataset from the Indian market together with a YOLOv5-based severity pipeline (when published, YOLOv5 itself preceded the YOLOv8 software release discussed in Section V-F). The three-quarter-view dataset [17] is a severity-classification resource for whole-car-side photos. A multi-view detection model (MVA-CDD, Peng et al., 2025) [20] confirms active research interest in multi-view fusion, which is beyond this project’s single-photo scope.

D. Dataset-family synthesis

Four dataset families recur: (i) real, segmentation-annotated (CarDD, VehiDE) — verifiable and licence-clear, the only suitable training source for this capstone; (ii) synthetic/procedural with perfect masks (CrashCar101) — a training complement, not validation evidence; (iii) severity classification (three-quarter-view; small severity sets); and (iv) closed commercial (unreleased insurer data, not verifiable).

E. Why CarDD is the working dataset

Two properties make CarDD [2] the defensible working dataset for a bounded capstone with a 4 GB GPU. First, it is verifiable: the dataset and its splits are public, and its annotation schema (polygon instance masks over six damage classes) can be audited locally; the AutoInspect-X local audit confirms train 2,816/ 6,211 instances, val 810/ 1,744, test 374/ 785. These instance counts document a further subtlety: the dataset is class imbalanced both at image level (dent and scratch predominate) and pixel level, so the metrics of (1)–(2) are reported per class as well as aggregated. Second, it is representative of the regime the demo targets: uncontrolled photographs with thin, spread-out, partly occluded damage. VehiDE is a natural future extension for cross-dataset generalization checks [3], [15], and CrashCar101 [4] could support pre-training or augmentation — both are explicitly deferred rather than adopted, because training on additional data without a documented, licence-reviewed protocol would violate the reproducibility rules of the project. Synthetic masks, in particular, are a complement for training but never validation evidence [4].

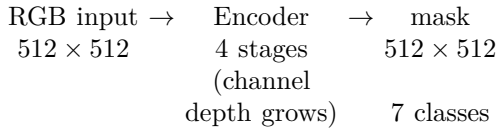


Fig. 1. Broad dataflow of the AutoInspect-X segmentation models: a fully-convolutional encoder-decoder with a bottleneck, evaluated by (1)–(2).

F. Small-damage framing across datasets

All three primary datasets are dominated by small damage instances in realistic photos — a localized dent or a scratch occupies a tiny fraction of the frame. The AutoInspect-X protocol therefore treats the small-damage slice as the binding constraint (Section III, warning 3) rather than as an afterthought, and reports slice metrics separately from aggregate metrics. This is an explicitly-stated project convention; the literature itself does not agree on a standard small-damage definition [1].

V. Convolutional Architectures for Segmentation

A. Fully convolutional networks and ResNets

Semantic segmentation in deep learning begins with the fully convolutional network (FCN), which replaces the fully-connected layers of a classifier with transposed-convolution upsampling so the network is trainable end-to-end on dense labels [21]. Backbone capacity is typically provided by residual networks (ResNet), whose identity skip connections enable very deep networks [22]. Both are the canonical building blocks of damage-segmentation pipelines: a classification backbone that has been turned into a dense-label decoder.

B. FCN and ResNet in the encoder-decoder pipeline

C. U-Net

The U-Net [5] is the canonical small-data segmentation architecture: an encoder that progressively downsamples while growing channel depth, a bottleneck, and a decoder that upsamples back to the input resolution, with skip connections carrying high-resolution detail from encoder to decoder. In its original formulation each stage is a pair of 3×3 convolutions (double-conv blocks), downsampling is max-pool, and upsampling is a transposed convolution; the final layer is a 1×1 convolution mapping features to logits per pixel. It was designed for biomedical data with limited labels and has become the de-facto baseline for damage segmentation because damage photos share that regime: few images, small and thin structures of interest. The AutoInspect-X baseline (ResNet34UNet, and its earlier CarddUNet predecessor) is a direct descendant of this design: the encoder stacks ResNet-derived blocks, and the decoder mirrors them with skip connections that concatenate encoder features to each upsampling stage.

D. Atrous designs and high-resolution representations

DeepLab-style designs generalize convolutions with atrous (dilated) rates to control the resolution of feature maps and capture multi-scale context without losing spatial detail [23]. HRNet instead maintains high-resolution representations throughout the network by fusing parallel multi-resolution branches, explicitly preserving the spatial detail needed for thin, spread-out damage [24]. For efficiency (relevant to the 4 GB GPU constraint of this capstone), MobileNetV2-style inverted-residual bottlenecks reduce FLOPs and parameters while retaining representational capacity [25].

E. Instance segmentation: Mask R-CNN

Mask R-CNN extends the Faster-RCNN detection paradigm with a parallel mask branch, producing an instance mask per detected object [18]. It is the reference model of the CarDD paper [2] and the standard instance-level baseline in damage work. Instance masks separate individual dents or scratches; semantic labels say nothing about which instance a pixel belongs to. For damage severity and counting, instance-level information is useful, but it requires bounding-box supervision and is heavier than pure semantic segmentation.

F. Detection frameworks used in applied damage work

Applied damage pipelines frequently use YOLO-family detectors. Ultralytics YOLOv8 [26] is open-source detection and segmentation software released in January 2023; it has no peer-reviewed paper and must be cited as a software release. YOLO-style dense detectors are typically trained with a focal loss that down-weights easy negatives, addressing the extreme background dominance of dense detection [27]. The focal loss

$$\text{FL}(p_t) = -\alpha(1 - p_t)^\gamma \log(p_t) \quad (3)$$

modulates the standard cross-entropy term by a factor $(1 - p_t)^\gamma$ that shrinks the gradient contribution of well-classified examples ($p_t \rightarrow 1$) and focusses training on hard, rare foreground pixels; the balance weight α handles foreground/background class imbalance [27]. In AutoInspect-X, a raw-PyTorch U-Net/hybrid is preferred over a YOLO pipeline to keep the loss, metrics, and class decoding under control on identical splits and seeds; a focal-loss variant is a listed alternative that has not been adopted, and its use would be recorded in an experiment decision rather than assumed.

G. Small-damage-specific considerations

Two mechanisms from the reviewed CNN literature are especially relevant to thin, spread-out damage: atrous (dilated) convolution and high-resolution fusion. Atrous convolution inserts dilation into the kernel so that a fixed-size filter covers a larger receptive field without extra parameters, letting multi-scale context be captured at full resolution [23]. HRNet carries a high-resolution branch

TABLE II

Representative architectures and their role in a damage-segmentation study. Architectural properties are stated as in the cited works; no benchmark comparison is implied.

Family	Representative	Relevant property
CNN	FCN [21]; ResNet [22]	Dense-label decoding; identity skips
Encoder–decoder CNN	U-Net [5]	Skips preserve fine detail; canonical baseline
Multi-scale (atrous)	DeepLab [23]; HRNet [24]	Atrous dilation; high-res fusion
Instance	Mask R-CNN [18]	Per-instance masks; CarDD’s reference model
Detection	YOLOv8 [26]; focal loss [27]	Applied pipelines; imbalance handling
Transformer encoder	ViT [6]; SETR [7]	Global context; patch sequences
Hierarchical transformer	Swin [10]; SegFormer [8]	Multi-scale; linearized attention
Hybrid	TransUNet [9]	Bottleneck attention + CNN decoder
Damage-specific	Mask R-CNN on CarDD [2]; ALBERT [19]; C-DiffDet+ [28]	Domain adaptations

through the whole network and fuses parallel streams, explicitly preserving the spatial detail that thin scratches and hairline cracks occupy [24]. Both ideas address the same failure mode that dominates under-trained damage models: the model collapses small structures into background. In the AutoInspect-X protocol this failure is measured directly by the small-damage slice (Section III, warning 3).

H. CNNs, transformers, hybrids: position in this review

I. CNNs, transformers, hybrids: role in this study

The baseline of AutoInspect-X is a U-Net-style encoder–decoder; the proposed model is a hybrid that inserts a bottleneck transformer. Both are trained with softmax cross-entropy over the argmax class target, on identical splits, seed, and schedule, and evaluated by (1)–(2). No cross-model comparative claim is drawn from other papers in this review.

VI. Attention and Global Context

Self-attention underlies the modern global-context family. The transformer architecture [29] computes pairwise relevance between all positions of a sequence, enabling direct long-range dependencies. Scaled dot-product attention computes, for query $\mathbf{Q}$, key $\mathbf{K}$, and value $\mathbf{V}$ matrices,

$$\text{Attention}(\mathbf{Q}, \mathbf{K}, \mathbf{V}) = \text{softmax}\left(\frac{\mathbf{Q}\mathbf{K}^\top}{\sqrt{d_k}}\right) \mathbf{V} \quad (4)$$

where d_k is the key dimension; the $\sqrt{d_k}$ scaling stabilizes the softmax as sequences grow. Multi-head attention runs this in parallel across several subspaces, allowing the model to attend to different relations simultaneously

[29]. For images, non-local networks graft a self-attention-style block onto conventional CNNs to capture long-range dependencies [30]. The Vision Transformer (ViT) [6] applies a standard transformer encoder directly to image patches, treating an image as a sequence of patch tokens and relying on large-scale pre-training for competitive accuracy.

Two properties matter for damage segmentation. First, attention is quadratic in the number of tokens: at $H \times W$ spatial resolution the score matrix $\mathbf{Q}\mathbf{K}^\top$ has $(HW)^2$ entries, which is why applying global attention to a high-resolution feature map is prohibitively expensive. Second, attention is a mechanism for context, not an architecture: it can be a whole encoder (ViT), a localized window (Swin) [10], or a bottleneck module inside a CNN. Where to place it is therefore a compute–accuracy trade-off, detailed next.

VII. Vision Transformers for Segmentation

A. ViT as an encoder: SETR

SETR [7] treats semantic segmentation as a sequence-to-sequence problem: a ViT encoder consumes the whole image as a sequence of patches, and a decoder upsamples to full resolution. This gives global context but increases memory and requires careful training; pure-encoder ViTs also lose the multi-scale feature hierarchy that CNNs provide for free.

B. Hierarchical transformers: SegFormer and Swin

Hierarchical transformers — Swin [10] with shifted windows and SegFormer [8] with a positional-encoding-free lightweight decoder — restore a CNN-like multi-scale pyramid. Swin restricts global attention to non-overlapping local windows and shifts the window grid between successive blocks so cross-window relations can propagate layer by layer; this replaces the quadratic all-pairs cost of (4) with cost that is linear in the number of patches. SegFormer uses overlapping patch merging, length-agnostic position encoding, and an all-MLP decoder, unifying semantic and panoptic segmentation. SegFormer was nominated as a candidate vision backbone in earlier AutoInspect-X documentation. The trade-off of both hierarchies is realized cost: a full hierarchical transformer encoder at high resolution is heavy for a 4 GB GPU, which motivates the hybrid designs below.

C. Hybrids as the pragmatic middle ground

For a bounded budget, pure-encoder and full-hierarchy definitions bookend the design space: a full ViT encoder maximizes context at high cost and loses multi-scale features; a plain CNN minimizes cost with local receptive fields only. CNN–transformer hybrids split the difference — the CNN keeps the high-resolution hierarchy and the transformer adds global context at a single, cheap resolution. This is the family the next section examines.

VIII. CNN–Transformer Hybrid Segmentation

A. TransUNet

TransUNet [9] is the direct methodological antecedent of the AutoInspect-X hybrid. It keeps a CNN encoder for high-resolution features, adds a transformer as a self-attention stage that operates only at the bottleneck (the most downsampled feature map), and then decodes with the original skip connections. This is a low-risk, GPU-bounded upgrade: global attention is applied where its resolution cost is smallest, while the CNN decoder and skips preserve fine detail. The hybrid design of this project (CarddHybrid, now superseded by the architecture-spec v3 HybridSegmentation) follows exactly this template.

B. Why bottleneck attention for damage

Practical constraints support the bottleneck location over a full ViT encoder for this capstone: (a) VRAM budget — the observed peak is about 2.85 GB for the baseline at base 64, and the added transformer capacity must stay within the 4 GB budget; (b) the CNN decoder’s skip connections remain necessary to recover small-damage detail; and (c) the transformer needs only the bottleneck’s spatial resolution (e.g., 32×32) to model inter-patch attention, whereas a full-patch ViT at 224×224 costs far more for the same end goal [6]. These are engineering constraints from the project’s own measurements, not general claims about which location is “best”.

C. Bottleneck attention in the AutoInspect-X models

In the architecture-spec v3 models, the transformer is a compact encoder stage applied to the 32×32 bottleneck feature map: sinusoidal positional encodings are added to the patch tokens, then residual self-attention layers (the head count and depth are fixed in the locked configuration) let tokens mix globally, and the result is projected back to a CNN decoder. Because the bottleneck token count is small ($32 \times 32 = 1024$ tokens), the quadratic cost of (4) stays well within the 4 GB budget. The decoder then upsamples with the original skip connections. This is the TransUNet pattern [9] adapted to damage segmentation, and it is the design whose preliminary pilots are reported in Section XIII.

D. Transformer-based automotive segmentation

Applying transformers directly to automotive damage, ALBERT [19] proposes a bidirectional-encoder-based instance segmentation over a large annotated automotive dataset (26 real damage types, 7 fake-damage artifacts, 61 distinct car parts). It is notable for explicitly disambiguating fake/ non-damage artifacts, a real-world nuisance in claim photos. The diffusion-plus- detector framework C-DiffDet+ [28] combines generative inpainting for anomaly scoring with a detection stage and a severity classifier, confirming a 2024–2025 trend of generative models converging with task-specific detectors.

IX. Uncertainty, Confidence, and Calibration

A. Types of uncertainty

Kendall and Gal [11] introduced the influential distinction between aleatoric (data-dependent, irreducible) and epistemic (model-dependent, reducible) uncertainty in vision. This taxonomy is the vocabulary of the AutoInspect-X honesty contract: a low-confidence flag must read as “the model itself is uncertain” (epistemic), not as evidence about the vehicle.

B. Practical estimators

Two practical epistemic estimators are established: MC dropout [12], which keeps dropout active at inference and averages stochastic forward passes as an approximate Bayesian marginalization; and deep ensembles [13], which train independently-seeded networks and use the variance across members as a predictive uncertainty estimate. Both carry a compute price — T stochastic forward passes or M networks respectively — which is significant on a 4 GB GPU and at demo-time inference. MC dropout is comparatively cheap to add to an existing checkpoint (dropout must already exist in the network); deep ensembles require multiple training runs, which is exactly the multi-seed protocol already planned for the final AutoInspect-X experiment. Both remain extensions rather than features of the shipped demo. Independent of the estimator, the integrity-relevant property is calibration: claimed confidence should match observed accuracy. The systematic review [1] notes that commercial damage tools do not publish calibration curves; a system that reports softmax mean confidence is only honest if it also flags low confidence loudly.

C. The honesty contract

AutoInspect-X implements a simpler, deterministic honesty signal — softmax mean confidence plus a low_confidence flag below thresholds — rather than MC dropout or ensembles, with the stated reason that the demo-grade baseline must flag itself honestly and often rather than emit confident-looking numbers. The flag is deliberately conservative: thresholds (min_mean_confidence, min_damage_fraction) are applied to the deterministic forward pass, and the engine labels every mask as MODEL PREDICTION even when confidence is high. MC dropout and deep ensembles remain listed extensions and would be adopted only via a documented decision. The proposed operational definition of the research question on confidence honesty is the separation between the confidence distribution over images where the predicted mask agrees with CarDD ground truth and the distribution where it does not.

D. Calibration as the integrity property

Regardless of the uncertainty estimator, the integrity-relevant property is calibration: the model is calibrated

when its reported confidence matches its observed accuracy, e.g., images predicted at 90% mean confidence are correct about 90% of the time. Softmax probabilities are well-known to be over-confident for out-of-distribution inputs, so reporting raw softmax confidence without a calibration check can mislead a user into treating a guess as a fact. None of the commercial damage tools surveyed in the systematic review [1] publish calibration curves; the honesty contract therefore treats calibration as a measured, reported property rather than an assumed one.

X. Research Gaps

Triangulating the reviewed evidence supports three gaps [1]–[3], [19]:

- 1) Detection and classification are saturated. The applied literature on damage detection/classification is large; commercial systems are closed-source, and calibration is under-reported.
- 2) Segmentation is the active front. CarDD, VehiDE, CrashCar101, ALBERT, and C-DiffDet+ all indicate that pixel-level and part-level segmentation is where the field is moving [2]–[4], [19], [28].
- 3) Confidence honesty is an under-published differentiator. The defensible, data-real opening is segmentation quality plus a verifiable confidence honesty contract, because public damage datasets supply masks but not financial or vehicle metadata ground truth. Cost prediction, in particular, depends on private datasets or synthetic price rules — neither is reproducible truth.

These gaps jointly argue against the two obvious temptations in automotive damage work. The first is to claim severity or cost: neither CarDD nor any verified public source provides the repair-cost or part-mask ground truth needed for an honest cost model. The second is to claim physical extent: an uncontrolled photograph has no scale reference, so physical area in cm^2 is not defensible; only a normalized image-relative damage-area ratio is. Both constraints come from the data, not the method, and they define the boundary of the honesty contract.

XI. Design Implications for the Capstone

The reviewed evidence translates into a small number of concrete design choices for AutoInspect-X, each traceable to a cited source:

- 1) Pixel-level task, foreground metrics. Segmentation answers “where and how extensive” [1], [2]; foreground-only mIoU and Dice avoid background inflation (Section III).
- 2) Shared baseline and protocol. A U-Net-style encoder–decoder [5] is the baseline; the hybrid adds a bottleneck transformer following TransUNet [9]. Identical splits, seed, schedule, loss, and metric harness for both arms is the only honest way to isolate the architecture effect [1].

TABLE III

Verified literature matrix: representative works, task, and the claim this review uses. Rows beyond the sourced claim are the review’s own integration, not claims made by the sources.

Work	Task / method	Claim used here (sourced/verified)
CarDD [2]	detection + instance seg	First public large-scale car-damage seg dataset; 6 classes; 4000 img / 9k+ inst
VehiDE [3], [15]	seg + detection + SOD	Public large-scale damage seg dataset; 13,945 img / 32k+ inst / 8 classes
CrashCar101 [4]	part + damage seg	Synthetic masks; complement, not validation
CDA-Net [16]	detection + severity	Applied YOLOv5 severity pipeline; 11,380-img damage set
MVA-CDD [20]	multi-view detection	Active multi-view damage research (out of single-photo scope)
Three-quarter view [17]	severity classification	Public severity/classification damage resource
C-DiffDet+ [28]	anomaly + detector + severity	2024–2025 generative + detector trend
ALBERT [19]	transformer instance seg	Transformer automotive seg with fake-damage disambiguation
U-Net [5]	encoder–decoder seg	Canonical small-data architecture; basis of the baseline
DeepLab [23]	seg with atrous conv	Multi-scale context at full resolution
Mask R-CNN [18]	instance seg	CarDD paper’s reference model
HRNet [24]	high-res multi-branch	Spatial-detail-preserving backbone
SegFormer [8]	hierarchical transformer seg	Position-free decoder; proposed vision backbone
TransUNet [9]	CNN + transformer hybrid	Bottleneck transformer; the hybrid’s antecedent
Kendall & Gal [11]	uncertainty taxonomy	Aleatoric vs. epistemic vocabulary of the honesty contract
MC dropout [12]	epistemic UQ	Test-time dropout uncertainty approximation
Deep ensembles [13]	epistemic UQ	Ensemble-variance uncertainty
Hasan et al. [1]	systematic review	Confirms saturated detection/classification; rare calibration

- 3) Attention cost is bounded. Global attention at the bottleneck (4) costs $O((HW)^2)$ in tokens, which is affordable at 32×32 but not at full resolution; the CNN decoder and skips preserve the small-damage detail (Sections V, ??).
- 4) Imbalance handling. Sparse-class damage needs either class re-weighting or a focal-style loss (3) [27]; whichever is chosen must be recorded as a decision, not defaulted silently.

- 5) Honesty is a measured property. Confidence is reported with a low-confidence flag and the mask is labelled MODEL PREDICTION; calibration and separation are planned measurement targets, not assumptions [11]–[13].
- 6) Reproducibility of records. Every run records experiment ID, dataset version and split, seed, code and model version, hyperparameters, metrics, and assumptions — consistent with the reproducibility expectations in the wider damage literature [1].

These choices are recommendations of this review; where AutoInspect-X has departed from a cited practice (for example, preferring a raw-PyTorch model over a YOLO pipeline, or reporting a deterministic honesty flag rather than MC dropout), the departure is a documented decision in the companion engineering documentation, not an accident.

A. Parameter budgets and the 4 GB training constraint

Efficiency matters in this capstone for a concrete and non-negotiable reason: the training and demo hardware is a 4 GB GPU. The Reviewed literature supplies two complementary responses. Architecturally, lightweight designs — MobileNetV2-style inverted residuals [25] — trade a small amount of accuracy for a large reduction in FLOPs and parameters, and hierarchical transformers with windowed attention [10] reduce the quadratic attention cost. Procedurally, the budget is respected by placing attention only at the bottleneck and by keeping the input at 512×512 for both arms. The measured engines of the two 15-epoch pilots are within this budget: the baseline defined and trained to roughly 24.6M parameters and the hybrid to roughly 27.9M parameters (not counts to be compared against any external checkpoint, but the two are directly comparable because they share the harness). The observed training peak for the earlier base-64 baseline was about 2.85 GB at batch 2; the pilot configuration is run within the same memory ceiling. Reporting these numbers is a reproducibility requirement [1], not an architecture claim.

XII. Threats to Validation in the Reviewed Literature

Three validation pitfalls recur across the works surveyed here and are worth naming explicitly because they are easy to fall into and hard to detect in a short review. Split leakage. Several damage studies do not state whether near-duplicate images of the same vehicle appear across train and test; the systematic review flags dataset provenance as a frequent weakness [1]. Benchmark incomparability. Papers report numbers on different dataset versions, crops, and evaluation code; treating them as a leaderboard is unsound. Overclaiming extent. Some applied work reports damage area or severity as if an uncontrolled photograph were a calibrated measurement; without a scale reference or calibration geometry this is not defensible. AutoInspect-X treats each of these as a documented protocol constraint rather than a review finding.

A. Verification discipline for research-facing claims

Two further disciplines apply because this review feeds a project that operates under adversarial research-integrity rules. First, the ground-truth policy of the project distinguishes REAL GROUND TRUTH (measured/verified by a trusted source, e.g., CarDD masks), WEAK LABEL (noisy/indirect), SYNTHETIC LABEL (rule/model-generated), DERIVED FEATURE (computed), MODEL PREDICTION (model output), and ASSUMPTION (team choice); each claim in this document, and each metric reported by the project, falls into exactly one of these categories. This is what stops a softmax score in a demo screenshot from being read as verified damage evidence, and what keeps the 15-epoch pilot numbers in Section XIII from being read as a research conclusion.

B. Threats to validation in this review

The review itself has three limitations. First, it is a targeted review for a bounded capstone, not an exhaustive survey; the systematic review of Hasan et al. [1] should be consulted for coverage. Second, architectural claims (“bottleneck attention adds global context”) are stated as the mechanisms described by the cited papers, not as benchmark accomplishments; no inter-paper comparison is drawn. Third, where the project’s own measurements are reported (parameter counts, VRAM use, pilot metrics), they are single-run observations from one environment and are explicitly not generalized. These limitations are intentional and keep the review within what the evidence supports.

XIII. Preliminary Experimental Observations (AutoInspect-X)

The following subsection is not a literature finding; it is the capstone’s own preliminary experimental observation and is included only so that the reader can relate the review to the companion architecture report. It is explicitly not a conclusion, and no statistical significance is claimed.

The project migrated from the demo-era CarddHybrid to the architecture-spec v3 research models: a baseline ResNet34UNet and the proposed HybridSegmentation (CNN encoder $\rightarrow$ transformer bottleneck $\rightarrow$ CNN decoder with skips). Two 15-epoch preliminary validation pilots (seed 0, full official CarDD splits, identical schedule, softmax cross-entropy over the argmax class target, EMA weights) were run as an intermediate check:

- pilot15_baseline: foreground mIoU 0.6127, foreground mDice 0.7440, pixel accuracy 0.8979 (best at epoch 14);
- pilot15_hybrid: foreground mIoU 0.5963, foreground mDice 0.7320, pixel accuracy 0.8873 (best at epoch 14).

Both models were still improving at epoch 14, and neither result establishes that either architecture is superior:

there is a single seed (0), no statistical test, and no fine-tuned schedule. The planned controlled comparison is a 60-epoch, 3-seed run under the locked protocol, which is required before any architecture-level claim. The demo default is `pilot15_hybrid` per the project plan, with `pilot15_baseline` retained as the controlled baseline arm. These pilots validate only that both model families train and evaluate end-to-end on CarDD under the locked schedule; they are recorded in the project registry together with their hyperparameters and seeds for reproducibility.

A. Pre-processing and harness

The AutoInspect-X harness pre-processes CarDD consistently: images are resized/cropped to 512×512 RGB, and the instance masks are collapsed to a per-pixel class label (background plus the six damage classes), preserving the official splits. Training applies horizontal flip and mild brightness/contrast augmentation only; validation and test are unaugmented. The two model arms share the identical loss (softmax cross-entropy over the argmax class target), the Adam optimizer, warm-up, cosine decay, EMA weights, and checkpoint rule (best validation foreground mIoU). Recording this protocol verbatim is what makes any future claim about one arm versus the other fair [1]; the pilots reported in Section XIII both used it.

B. What the small-damage slice shows

The small-damage slice defined in Section III is not only a metric convention; it is where the practical difficulty concentrates. In the CarDD training set, the 25th-percentile instance-area threshold is about 3,013. pixels at 512×512 (i.e., roughly 1.1% of the image), a measured baseline against which “small” is defined in the project protocol; instances at or below this threshold are overwhelmingly dents and scratches — the two most common damage classes in real claims. Because these classes occur at tens of pixels in width, a model trained with a plain cross-entropy objective can under-segment them entirely, collapsing them into background. The pilot runs do not yet claim to have fixed this: the small-damage slice remains the binding constraint on trust in both architectures, and it is reported explicitly rather than hidden in the aggregate. This measurement is project-specific and is included to illustrate why the review repeatedly stresses fine-structure recovery mechanisms (U-Net skips, atrous context, high-resolution fusion) rather than as a benchmark claim.

XIV. Conclusion

Vehicle damage segmentation is a reproducible, data-bounded research problem. Real, public, mask-annotated datasets exist (CarDD [2], VehiDE [3]), synthetic resources (CrashCar101 [4]) complement them for training, and calibration/confidence honesty is under-published despite being the natural integrity constraint on a photo-first inspection system. Architecturally, the evidence supports a CNN–transformer hybrid as a GPU-bounded, contemporary upgrade to a U-Net baseline — attention placed at the

bottleneck preserves fine detail via the CNN decoder while adding global context. The AutoInspect-X pilots confirm only that the two model families train on CarDD under the locked schedule; the architecture comparison remains to be settled by the planned multi-seed experiment, and the value of the system hangs on its honesty contract, not on any single metric.

Three statements summarize the position this review leaves the reader with. (1) Task selection determines honesty. A segmentation task on a verified, licence-clear dataset (CarDD) is reproducible; severity/cost claims are not, because the required ground truth does not exist publicly. (2) Architecture is an affordable open question. The CNN–transformer hybrid family is contemporary and GPU-bounded; it is answerable in a capstone by a locked, multi-seed protocol — no other paper is needed to establish the comparison. (3) Confidence labelling is the under-exploited, measured contribution. Reporting a low-confidence flag and measuring agreement–disagreement separation converts “the model was confident” from a marketing sentence into a testable property [11]. Everything else in damage analysis worth doing remains constrained by data that has not yet been made public.

Acknowledgment

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